

Four-Way Yield Comparison

Actual measured yield, modelled predicted_Yield, and our recommended model, in plain terms

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Three yield numbers exist per tree, and this report compares them directly. **Actual yield** is the real, weighed harvest, the only ground truth we have. **Modelled predicted_Yield** is a number produced by an earlier model, built mostly from canopy and growth measurements, not a real weighing. **Our model** is a small model we built and tested that predicts yield from cultivar, bloom intensity (EBI), climate, and canopy growth. All comparisons below use the same 167 trees (2022 and 2023 combined), the ones where all three numbers exist.

This report is in two parts, both presented in full. The open question at the end of Part B is addressed directly to the lab.

Part A

Real, measured yield. The only ground truth available (n=167 tree-years).

46% vs. 29%

real yield variation explained: our model vs. predicted_Yield

Part B

Modelled predicted_Yield, on its own terms, over a larger sample (n=1793 tree-years).

84.4%

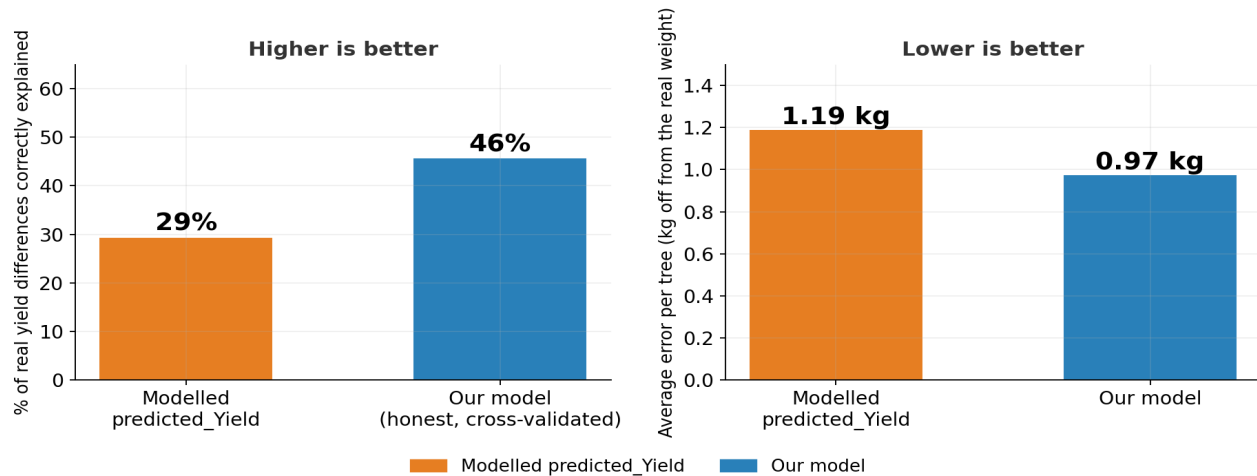
of predicted_Yield's own variation explained, remote-sensing-only

Part A: Real, Measured Yield

Checked against the actual, weighed harvest, the only ground truth in this dataset (n=167 tree-years, 2022 and 2023 combined).

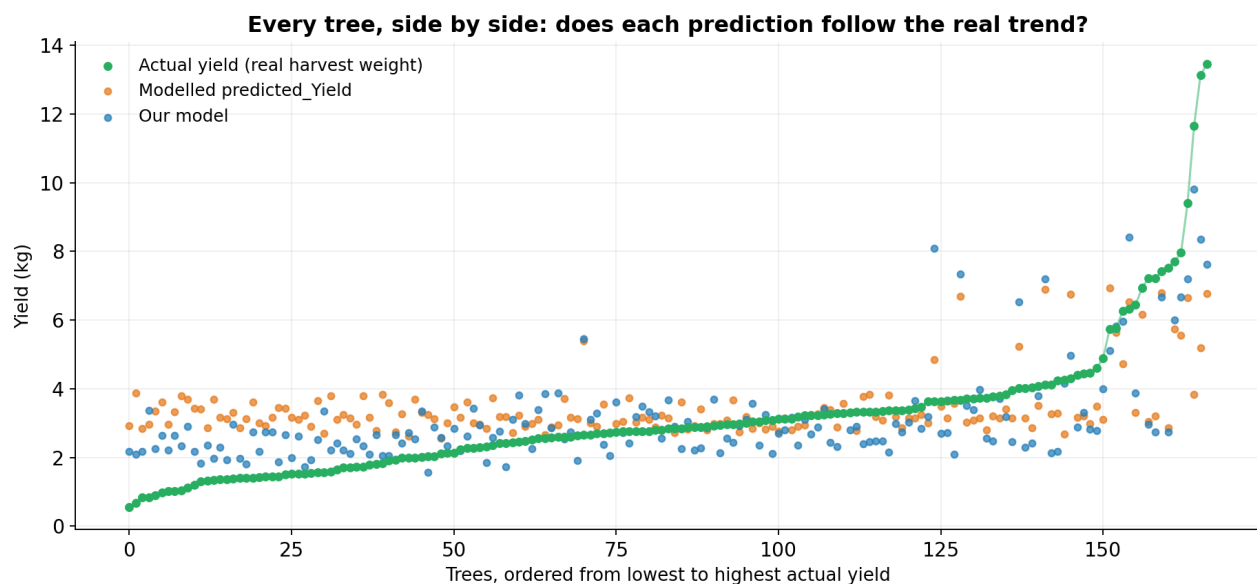
The big picture

The big picture: which prediction is closer to the real, weighed yield?



Left: out of all the differences between trees in real yield, our model correctly explains about 46%; predicted_Yield explains about 29%. Right: on average, our model's guess is off by 0.97 kg per tree, predicted_Yield's guess is off by 1.19 kg per tree (average real yield is about 3.16 kg). Our model is closer on both measures.

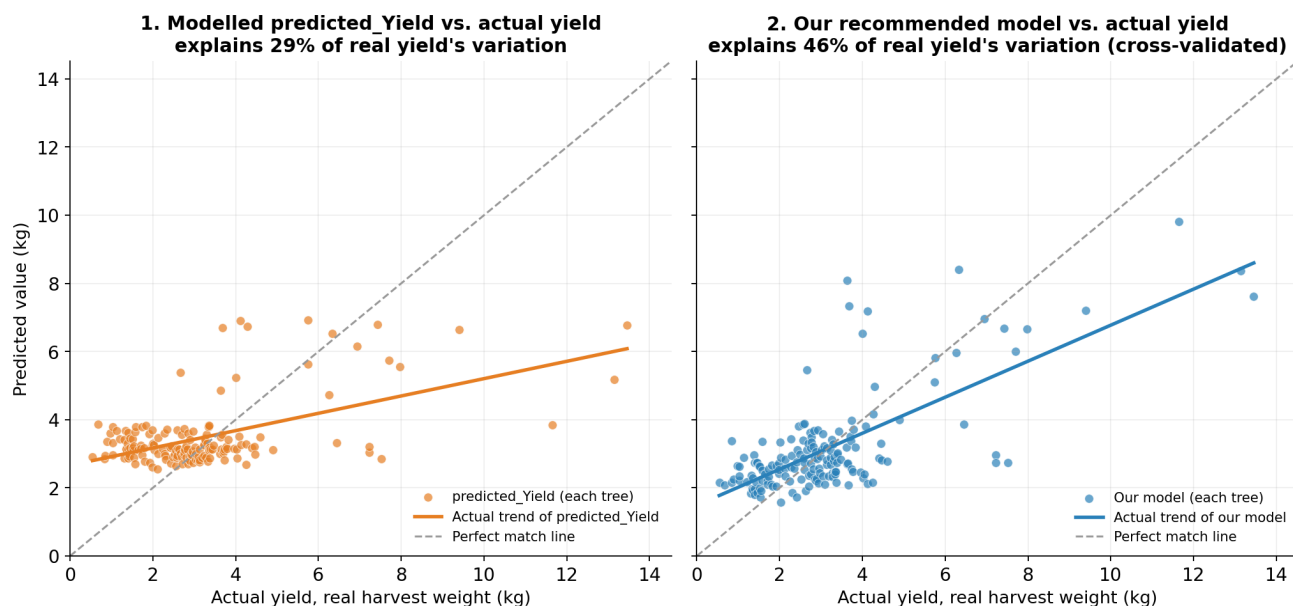
Every tree, side by side



Trees are lined up from lowest to highest real yield (green line). Orange dots (predicted_Yield) stay fairly flat and do not rise much even as real yield climbs, especially on the right side of the chart, the highest-yielding trees. Blue dots (our model) rise and fall more closely with the green line, particularly at the high-yield end.

1 & 2. Modelled predicted_Yield vs. actual yield, and our model vs. actual yield

Side by side: same real trees, same scale, two different predictions

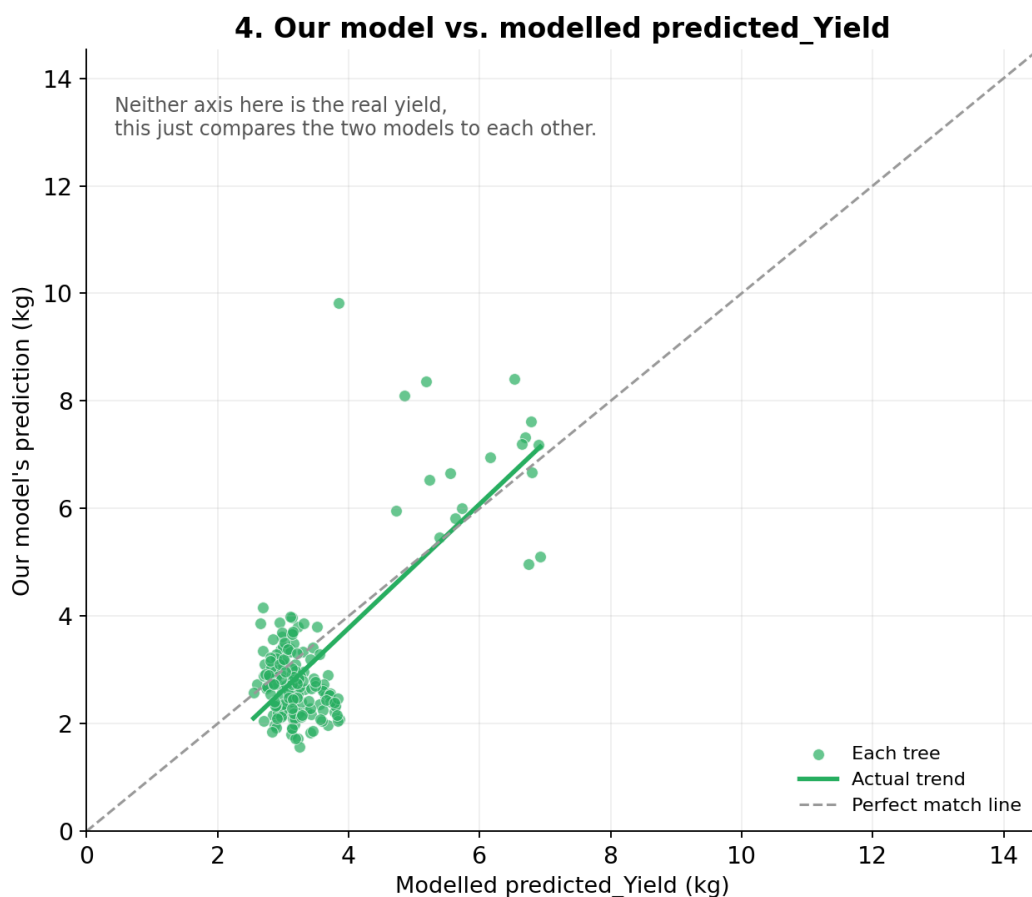


Same real trees, same axis scale, placed next to each other on purpose. Left: predicted_Yield vs. actual yield, $r=0.54$, explains 29% of real yield's variation ($n=167$). Right: our model vs. actual yield, $r=0.73$ on the data it was fit on (53% explained, $n=167$). In both panels, dots closer to the dashed 1:1 line are more accurate; the left panel's orange trend line sits visibly below the diagonal, the right panel's blue trend line tracks it more closely.

predicted_Yield can reconstruct itself almost perfectly (94 to 96%) when tested against its own ingredients, but here, against a real scale, it does much worse. That gap is the key finding: looking accurate on paper does not mean it predicts the real harvest well.

Note on our model's number: 53% (shown above) is how well it fits the same data it was built on, which is always a bit optimistic. The fairer, stricter test (fitting on some trees, checking on others it never saw) gives 46%, and that is the number to trust for how the model will do on a new tree. Even at 46%, it is still clearly ahead of predicted_Yield's 29%.

4. Our model vs. modelled predicted_Yield



$r=0.75$ ($n=167$). Here, neither number is the real yield, this just checks whether the two models agree with each other. They do, fairly closely, but that is expected rather than reassuring: both use cultivar and overlapping growth measurements as inputs, so part of the agreement simply comes from sharing ingredients. Two models agreeing with each other is not the same as either one being right about the real harvest, only the comparisons against actual yield (sections 1 and 2) establish that.

Bottom line (pooled, both cultivars combined)

Checked against the real, weighed yield, our model is clearly closer than predicted_Yield: it explains more of the real variation (46% vs. 29%) and its average error per tree is smaller (0.97 kg vs. 1.19 kg). predicted_Yield looks far more accurate on paper only because it is being asked to reproduce a number built from its own ingredients. Our model, though modest by CV standards, is the one actually tested against, and closer to, what really came off the trees.

Full statistical comparison

Comparison	r	R ²	MAE (kg)	p-value	n
predicted_Yield vs. actual yield	0.54	29%	1.19	<0.001	167
Our model vs. actual yield (fit to its own training data)	0.73	53%	0.97	<0.001	167
Our model vs. actual yield (cross-validated, the honest number)	n/a	46%	1.02	n/a	167
Our model vs. predicted_Yield (agreement only, not accuracy)	0.75	56%	n/a	<0.001	167

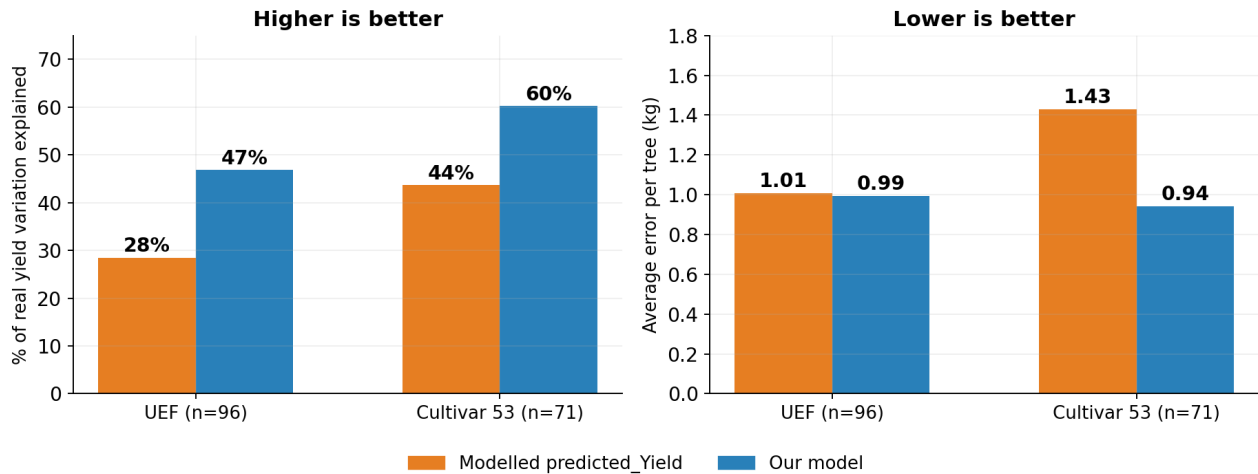
Why two R² numbers for our model: 53% is how well it fits the same data it was built on, always a bit optimistic. 46% is the stricter test, fit on some trees, checked on others it never saw, and is the number that predicts how the model will do on a new tree. Cross-validated MAE (1.02 kg) is likewise a touch higher than the in-sample MAE (0.97 kg) for the same reason; both are still well ahead of predicted_Yield's 1.19 kg.

Now split by cultivar

This orchard's own key finding is that cultivar changes the bloom-yield relationship, so every comparison above is repeated separately for UEF (n=96 tree-years) and cultivar 53 (n=71 tree-years).

The big picture, by cultivar

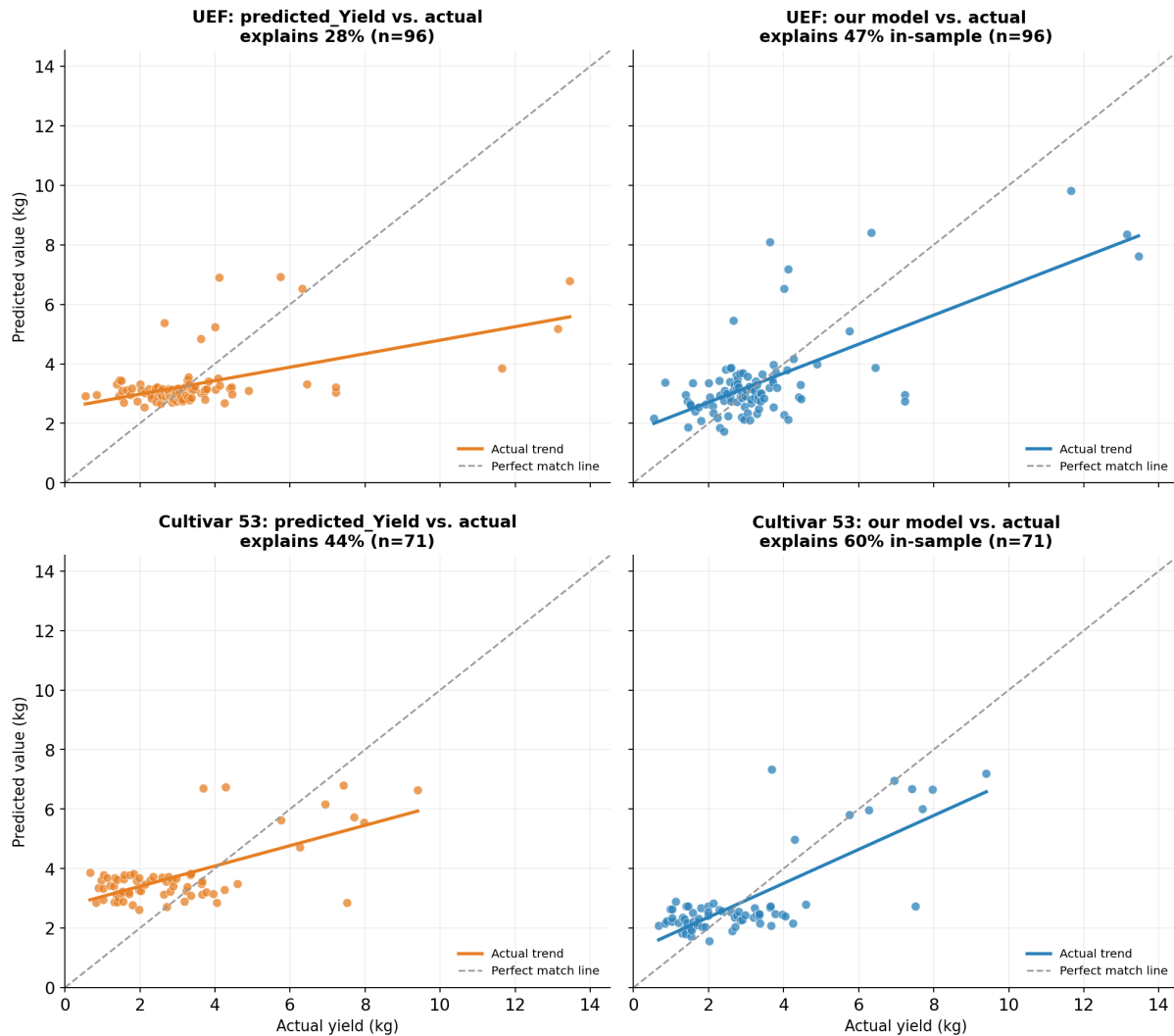
The big picture, split by cultivar



UEF: our model explains 47% of real yield vs. predicted_Yield's 28%, average error 0.99 kg vs. 1.01 kg. Cultivar 53: our model explains 60% vs. predicted_Yield's 44%, average error 0.94 kg vs. 1.43 kg. Our model is closer to actual yield in both cultivars, and the gap is largest in cultivar 53.

1 & 2. Predicted vs. actual, and our model vs. actual, by cultivar

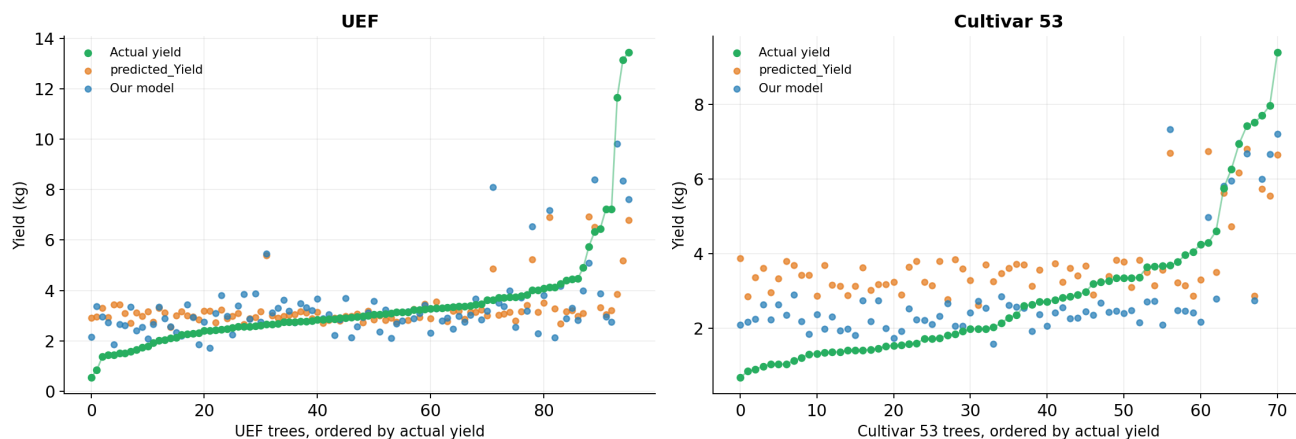
Predicted vs. actual, split by cultivar (each row is one cultivar)



Top row: UEF trees. Bottom row: cultivar 53 trees. Same axis scale within each row so the two columns are directly comparable. In both cultivars the right-hand column (our model) sits closer to the dashed 1:1 line than the left-hand column (predicted_Yield).

3. Every tree, by cultivar

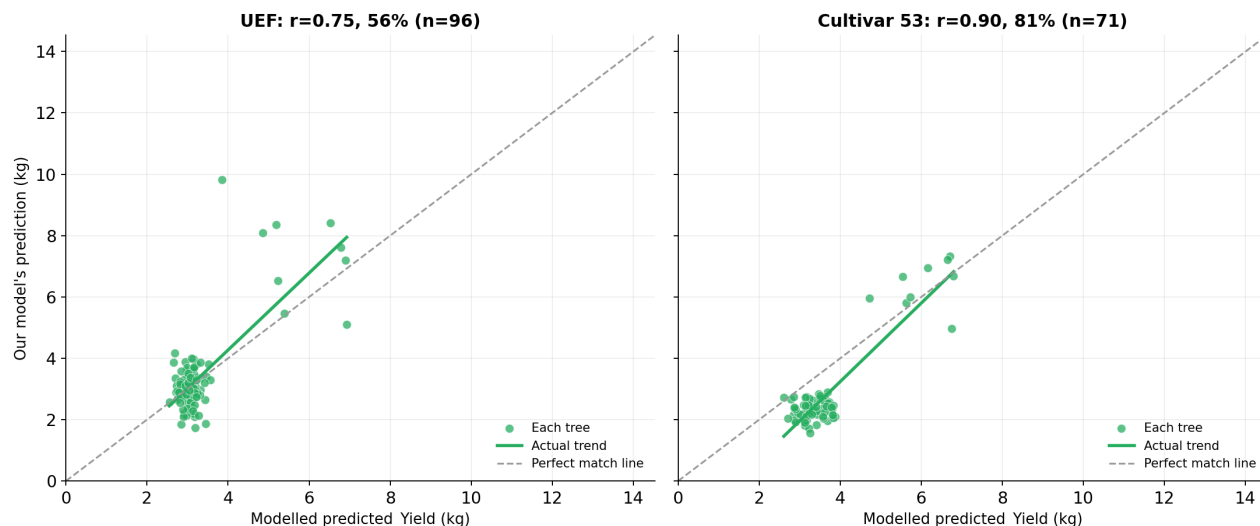
Every tree, side by side, split by cultivar



Same layout as the pooled version, one panel per cultivar, trees ordered by actual yield within their own cultivar. The blue (our model) points track the green (actual) line more closely than the orange (predicted_Yield) points in both panels.

4. Our model vs. modelled predicted_Yield, by cultivar

Our model vs. modelled predicted_Yield, split by cultivar



UEF: $r=0.75$ (56%, $n=96$). Cultivar 53: $r=0.90$ (81%, $n=71$), noticeably higher than UEF's. As with the pooled version, this is agreement between two model outputs, not validation against real yield, so a higher number here is not automatically good news. In cultivar 53 specifically, treat this stronger agreement with extra caution: $n=71$ is a modest sample to split further, and both models share cultivar and Growth_April as inputs, so part of an unusually high match can reflect shared ingredients and a smaller sample rather than either model being more correct about cultivar 53's real yield.

Bottom line, by cultivar

In both cultivars, our model tracks real, measured yield more closely than predicted_Yield does, on every measure checked: more variation explained, smaller average error, and a tighter visual fit to the 1:1 line. The gap between our model and predicted_Yield is wider in cultivar 53 than in UEF. This is directional at n=71 and n=96, not the fully independent, cross-validated per-cultivar test this project would run before finalizing a cultivar-specific claim; the pooled model's cross-validated $R^2=0.456$ remains the primary, honest number for this project.

Cultivar	n	Mean yield (kg)	R ² pred.	R ² ours	MAE pred. (kg)	MAE ours (kg)
UEF	96	3.38	28%	47%	1.01	0.99
Cultivar 53	71	2.85	44%	60%	1.43	0.94

Cultivar 53's MAE improvement (1.43 to 0.94 kg, a 34% cut) is the largest gain of any group in the orchard; UEF's variation-explained gain is large (28% to 47%) but its MAE is roughly flat, since UEF's predicted_Yield was already reasonably close on average error even though it explained less of the real spread.

Why this matters

The bloom imagery this model relies on is captured in February, five months before harvest. A forecast built from it is available for planning labor, storage, and logistics far earlier than harvest itself, and does not require waiting on a July acquisition the way some of the spectral indices tested elsewhere in this project do.

Because the bloom-yield relationship runs in opposite directions by cultivar, more bloom intensity predicts more yield in UEF ($r=+0.29$) but less yield in cultivar 53 ($r=-0.26$), a single orchard-wide average would miss this entirely. Splitting the model by cultivar is what lets it flag an underperforming block before the yield gap shows up at harvest, rather than after. The same drone-plus-model approach extends to any block with cultivar identity and bloom imagery, it is not specific to Plot A.

None of this changes the honest ceiling on the numbers above: 46% of real yield variation explained, cross-validated, is a genuine result and not a small one against a 29% baseline, but it is not a finished forecasting product. The single largest lever for improving it further is more measured-yield seasons, not new features or alternative models; fourteen other model families were tested against this one over the course of this project and none beat it.

A note on the trees that stand out: the highest- and lowest-yielding trees in this sample were checked as possible anomalies and, separately, fourteen alternative models (regularized, robust, and nonlinear) were tried against this one. Both checks agreed the extreme trees carry real, correctly-explained information rather than noise, and none of the alternative models beat this one, so all of them stay in every number above rather than being removed.

Part B: Modelled predicted_Yield

predicted_Yield treated on its own terms: which features best explain it, and how well, over a much larger sample (1793 tree-years, 2022 and 2023, versus 167 in Part A).

What best explains predicted_Yield

A separate question from Part A: not how well a model predicts the real harvest, but which features best explain the existing **predicted_Yield** number itself. This does not require the measured-yield CSV, so it runs on a much larger sample, every UEF and cultivar 53 tree with a predicted_Yield value in 2022 or 2023 (1793 tree-years, versus 167 in Part A).

Physiology fields known to be near-direct ingredients of predicted_Yield's own construction (SWP and Growth at April/May-June/June, CNC_June, and the cluster fields; CNC_June alone correlates $r=+0.95$ with predicted_Yield) were deliberately excluded from the search. Including them would not be a genuine feature search, it would just reconstruct the label from its own known ingredients. The candidate pool here is remote-sensing only: cultivar, bloom intensity (EBI) and its cultivar interaction, climate, canopy growth, plus the same 34-column engineered bank (canopy structure, EBI-pixel distribution, composite indices) used throughout this project.

Model	Features	R ² (CV)	AIC	MAE (kg)
Baseline (4 features)	cultivar, EBI×cultivar, climate, Growth_April	84.4%	2810	0.37
Best remote-sensing-only model (search over 34 candidates)	same 4, none of the 34 added	84.4%	2810	0.37
With SWP/Growth_May_June/ Growth_June/CNC_June allowed back in	+ CNC_June, Growth_June	93.5%	1249	0.26

Result: the four base features alone already explain 84% of predicted_Yield's variation, cross-validated, dramatically higher than the 46% ceiling for real measured yield above. A systematic add-one screen across all 34 engineered candidates, followed by forward selection, found nothing worth adding: the closest candidates were CanopyCover ($\Delta\text{AIC}=-11.2$, $\Delta\text{CV R}^2=+0.07$ pts); EBI_x_CanopyCover ($\Delta\text{AIC}=-7.4$, $\Delta\text{CV R}^2=+0.04$ pts); BrightFraction_x_CanopyArea ($\Delta\text{AIC}=-0.3$, $\Delta\text{CV R}^2=-0.02$ pts), all negligible in practical terms despite the large sample. The search added zero features; the 84% baseline stands as the best remote-sensing-only model found.

The bottom row of the table above shows what happens when the excluded physiology fields are allowed back in, exactly the check requested. Adding **CNC_June alone** jumps cross-validated R² from 84.4% to 93.2% in a single step ($\Delta\text{AIC}=-1478$, an enormous, not a marginal, improvement), and **Growth_June alone** would do nearly as much on its own (88.4%, $\Delta\text{AIC}=-537$). Forward selection with all fields available picks exactly these two and stops at 93.5% cross-validated R², AIC falling from 2810 to 1249 and MAE from 0.37 kg to 0.26 kg, far larger jumps than any of the 34 legitimate remote-sensing candidates produced individually or together.

This is a concrete illustration of the circularity concern noted above: CNC_June and Growth_June are close to the arithmetic the clustering model used to produce predicted_Yield in the first place, which is a plausible reason one field alone outperforms all 34 remote-sensing candidates combined. The 84.4% remote-sensing-only figure answers the main question of this part; the 93.5% figure with physiology fields included answers a different, narrower question, how well predicted_Yield can be reconstructed from fields close to its own construction, and is presented here for completeness rather than as the headline result.

Does Random Forest do any better?

Model	Features	R ² (CV)	vs OLS
OLS, remote-sensing-only	4 base features	84.4%	baseline
Random Forest, remote-sensing-only	same 4 base features	85.4%	+1.0 pts
Random Forest, + 34 engineered candidates	38 features, RF's own selection	85.4%	+1.0 pts
OLS, with physiology fields	+ CNC_June, Growth_June	93.5%	(see note)
Random Forest, with physiology fields	44 features, RF's own selection	98.7%	(see note)

Random Forest (same from-scratch implementation and tuning protocol used throughout this project, re-tuned for this much larger sample) was tested against both versions of this search. On the remote-sensing-only comparison, RF narrowly beats OLS: 85.4% versus 84.4% cross-validated R², a real but modest gain of about one point. Giving RF access to the other 34 engineered candidates does not add anything further (85.4%, statistically the same as the 4-feature result), which agrees with the OLS forward-selection finding above by a different method: there is nothing more in those 34 features for either a linear or a nonlinear model to find.

With the physiology fields allowed back in, cross-validated R² reaches 98.7%, well above OLS's 93.5%. CNC_June alone correlates $r=+0.95$ with predicted_Yield, and a flexible nonlinear model with most of its 44 features available on each split is well-positioned to fit that relationship closely. Read this as Random Forest reconstructing predicted_Yield from fields close to its own construction, rather than as a stronger model of the underlying phenomenon. As with the OLS comparison, the remote-sensing-only numbers are the primary ones for this part; the physiology-included numbers, OLS or RF, are included for completeness.

What the 34 engineered features (and the 4 base features) actually contribute (RF permutation importance)

An earlier version of this table used split-based (Gini-style) RF importance, the standard measure but one with a known bias: it rewards features that get split on often, which favors continuous features over discrete ones regardless of true effect size. That bias showed up concretely here: **climate** (Chill Portions) is station-level, exactly 2-valued in this sample (35 in 2022, 25 in 2023), the same cardinality as the binary cultivar indicator, yet split-based importance still ranked it far above cultivar. The reason turned out to be real, not an artifact of cardinality: the single available climate split separates predicted_Yield by 2.43 kg between years, versus only 0.21 kg between cultivars, so climate's one split is simply far more informative, not more frequent.

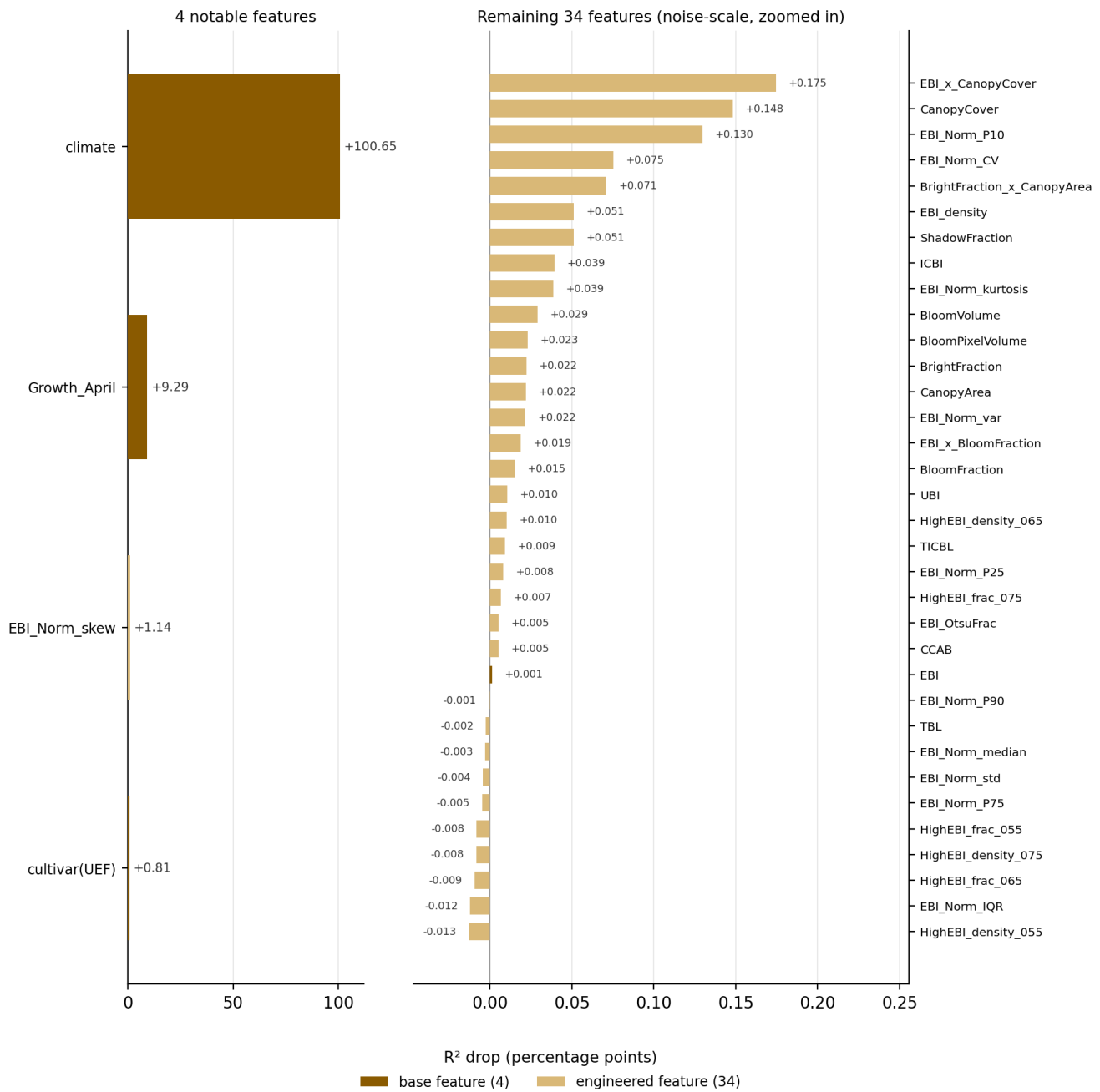
This table instead uses permutation importance: after training each cross-validation fold's forest, one feature at a time is shuffled within the held-out test fold and the drop in cross-validated R^2 is measured, using the same trees (no retraining). Averaged over 5 seeds x 5 folds x 25 trees per fold (baseline CV $R^2=85.4\%$, consistent with the 85.4% figure above). Unlike split-based importance, this is not biased by how many distinct values a feature has, only by how much the model's accuracy actually depends on it.

Result: cultivar looks completely different under this fairer measure. Shuffling **cultivar** costs 0.81 R^2 points, the 4th largest drop of all 38 features, comfortably above the engineered bank's best entry (EBI_Norm_skew, 1.14 points). Shuffling **climate** costs 101 points, effectively destroying the model, confirming it as the single dominant driver of predicted_Yield in this sample. Base **EBI** (mean bloom intensity) still contributes almost nothing on its own (+0.001 points, rank 28 of 38), consistent with the project's established finding that mean bloom level is flat and it is variance/tails and cultivar identity that carry the real signal for actual yield.

#	Feature	R ² drop	Description
1	climate	+100.648	Chill Portions (Dynamic Model), station-level: 35 in 2022, 25 in 2023.
2	Growth_April	+9.285	Canopy growth measured in April.
3	EBI_Norm_skew	+1.145	Skewness of the in-crown EBI distribution.
4	cultivar(UEF)	+0.813	Cultivar indicator (1 = UEF, 0 = cultivar 53).
5	EBI_x_CanopyCover	+0.175	EBI × canopy cover fraction.
6	CanopyCover	+0.148	Canopy pixels / (canopy + ground) footprint fraction.
7	EBI_Norm_P10	+0.130	10th percentile of per-pixel EBI.
8	EBI_Norm_CV	+0.075	Coefficient of variation of in-crown EBI (patchiness).
9	BrightFraction_x_CanopyArea	+0.071	BrightFraction × canopy area.
10	EBI_density	+0.051	EBI per unit canopy area.
11	ShadowFraction	+0.051	Fraction of the footprint classified as shadow (Otsu on brightness).
12	ICBI	+0.039	Illum.-Corrected Bloom Intensity: EBI × (1 – Shadow).
13	EBI_Norm_kurtosis	+0.039	Kurtosis of the in-crown EBI distribution.
14	BloomVolume	+0.029	EBI × canopy area (intensity × size proxy).
15	BloomPixelVolume	+0.023	BloomFraction × canopy pixel count.
16	BrightFraction	+0.022	Bright canopy pixels overall, not bloom-specific.
17	CanopyArea	+0.022	Canopy footprint area in m ² .
18	EBI_Norm_var	+0.022	Variance of per-pixel EBI in the crown.
19	EBI_x_BloomFraction	+0.019	EBI × BloomFraction.
20	BloomFraction	+0.015	Canopy pixels classified as bloom (Otsu on brightness).
21	UBI	+0.010	Uniform Bloom Intensity: EBI × (1 – EBI_Norm_CV).
22	HighEBI_density_065	+0.010	Pixels with EBI above 0.65, per m ² .
23	TICBL	+0.009	Total Illum.-Corrected Bloom Load: area × EBI × (1 – Shadow).
24	EBI_Norm_P25	+0.008	25th percentile of per-pixel EBI.
25	HighEBI_frac_075	+0.007	Fraction of pixels with EBI above 0.75.
26	EBI_OtsuFrac	+0.005	Fraction above a per-crown, EBI-specific Otsu threshold.
27	CCAB	+0.005	Canopy-Completeness-Adj. Bloom: HighEBI_frac_065 × CanopyCover.
28	EBI	+0.001	Mean EBI for the crown (overall bloom intensity).
29	EBI_Norm_P90	-0.001	90th percentile of per-pixel EBI.
30	TBL	-0.002	Total Bloom Load: canopy area × HighEBI_frac_065.
31	EBI_Norm_median	-0.003	Median per-pixel EBI in the crown.
32	EBI_Norm_std	-0.004	Std. dev. of per-pixel EBI in the crown.
33	EBI_Norm_P75	-0.005	75th percentile of per-pixel EBI.
34	HighEBI_frac_055	-0.008	Fraction of pixels with EBI above 0.55.
35	HighEBI_density_075	-0.008	Pixels with EBI above 0.75, per m ² .
36	HighEBI_frac_065	-0.009	Fraction of pixels with EBI above 0.65 (project's bloom-bright cutoff).
37	EBI_Norm_IQR	-0.012	Interquartile range (P75 – P25) of per-pixel EBI.
38	HighEBI_density_055	-0.013	Pixels with EBI above 0.55, per m ² .

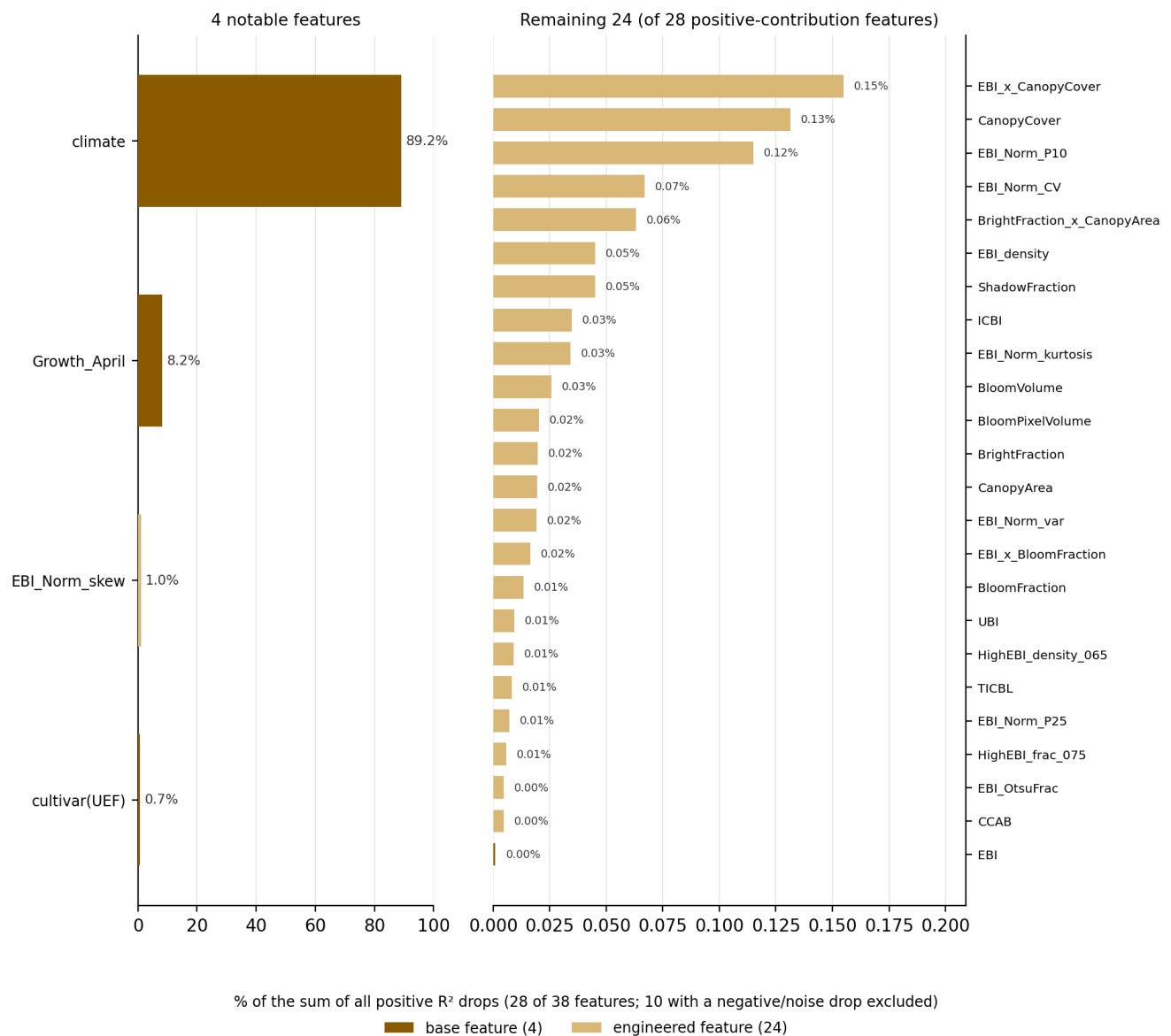
All 38 features (4 base, highlighted in amber, plus all 34 engineered candidates), ranked by permutation R² drop, highest first. Negative values mean shuffling that feature very slightly improved cross-validated accuracy, a sign of pure noise, not a real negative effect.

Permutation importance: R² drop by feature, all 38



Same 38 features and values as the table, as bars. Left: the 4 features with a real (non-noise) effect, linear scale. Right: the remaining 34 engineered features, zoomed into their much smaller range so their small relative differences are still visible.

Share of total positive contribution, by feature



The same result read as a share of the total: of the 28 features with any real positive contribution (the other 10 have a negative, noise-only drop and are excluded here), climate alone accounts for 89.2%, Growth_April 8.2%, EBI_Norm_skew 1.0%, and cultivar 0.7%; every one of the 34 engineered features individually accounts for well under 0.2%.

Why this matters for how to read predicted_Yield: it is not that remote sensing happens to explain the real harvest poorly (46%) but explains this other number well (84%) because it is a better hidden signal. It is that predicted_Yield is itself a smooth, model-generated number, mechanically close to a function of a few correlated growth, climate, and cultivar inputs, so a handful of features saturate it almost immediately. Real, weighed yield is noisier and does not saturate the same way. High internal consistency for predicted_Yield and modest cross-validated accuracy against the real harvest are two sides of the same finding already established in Part A, not a contradiction.

Open question for the lab: everything in this part follows from how predicted_Yield was built, not from an assumption that it is unreliable. If there is a validation of predicted_Yield against a larger, independent set of real measured-yield trees (sample size, correlation, method), sharing it would speak directly to the circularity concern raised above. Depending on what that validation shows, predicted_Yield could reasonably move from a modelled proxy to interpret with caution toward a usable stand-in for real yield, which would change which of the two parts of this report this project should build on going forward. We would welcome that data and would update this analysis accordingly.